

Andrey Zaytsev

📍 Lisbon, Portugal · Open to relocation across Europe

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Senior Product Security / Application Security Engineer with 10+ years in software and security engineering, including 5.5 years securing Chrome at Google. Expertise in threat modeling, security architecture and secure code review, web/API and identity security, vulnerability remediation, and secure SDLC/DevSecOps across CI/CD, containers, and AWS.

Experience

Independent Product Security Engineer

Jul 2025 - Present

- Designed and built TutorOps, a Java 21/Spring Boot 4.1 platform with security invariants and deny-by-default RBAC; implemented OAuth 2.0/OIDC Authorization Code + PKCE with Keycloak/Spring Security and validated JWT signature, issuer, audience, expiry, and roles
- Built GitHub Actions DevSecOps gates for Maven tests, CodeQL SAST, Dependabot/dependency review, Gitleaks, Trivy SCA/container/IaC scanning, and CycloneDX application and image SBOMs
- Hardened a non-root Docker runtime with a read-only filesystem and dropped Linux capabilities; traced and remediated a high-severity transitive CVE and made fixable high/critical findings release-blocking
- Deployed TutorOps on AWS EC2 with Terraform using encrypted EBS, IMDSv2, restricted AWS Security Group ingress, HTTPS termination, and no inbound SSH
- Built and operate solange.top, a production React/TypeScript product on Vercel with Stripe and Cal.com integrations

Chrome Privacy & Security Engineer

Google · Munich, Germany

Jan 2020 - Jun 2025

- Led the end-to-end remediation of a Chrome Android update-detection vulnerability that could suppress warnings for outdated, vulnerable browser versions; redesigned the client/server flow, fixed protocol and threading defects, and drove backend changes across Chrome Update infrastructure teams
- Owned Chrome Android Safety Check from technical design through launch as a prominent top-level Settings entry; obtained security, privacy, and accessibility approvals and kept leadership aligned while unifying compromised-password, Safe Browsing, and browser-update risks in one remediation workflow
- Owned 200+ end-to-end security/privacy launch reviews using data-flow tracing, trust-boundary/abuse-case analysis, architecture review, and threat modeling; drove revisions and escalated 3+ launch blocks, partnering with director-level leadership on mitigations or policy exceptions
- Selected as a Chrome telemetry approver, especially for Android; embedded reusable guidelines in the standard review workflow for 10+ teams, reducing repeat policy violations involving PII or linkable data and accelerating reviews
- Served as Chromium code owner and mobile-development point of contact, conducting 500+ C++/Java/JNI code reviews and surfacing pre-merge security and reliability risks in lifecycle handling, feature gating, API boundaries, and test coverage; mentored an engineer to independent ownership
- Analyzed Chrome Android and CI logs to debug cross-layer failures; in recurring on-call, restored builders, rolled back breaking changes, and helped owners fix and safely reland 20+ changes

Search Infrastructure & Reliability Engineer

Google · Chicago, IL, USA

Oct 2018 - Jan 2020

- Designed and operated Borg-based infrastructure (Google's Kubernetes precursor) and Monarch monitoring across 1,500+ service instances; defined SLAs and alerts and improved release-signal reliability by eliminating configuration drift and flaky infrastructure failures

Software Engineer

Jump Trading · Chicago, IL, USA

Jun 2016 - Jul 2018

- Built large-scale C++/CUDA and Splunk systems, including a 100M-event/day analytics pipeline that cut anomaly diagnosis from hours to minutes and a 500K-write/sec storage engine with sub-millisecond p99 latency

Software Engineering Internships

San Francisco Bay Area, USA

2014 - 2015

- Dropbox:** Shipped a Python/React/TypeScript distributed system for team-administration actions at 100,000-user scale
- Apple:** Improved iOS and OS X input-method usability and accessibility in Objective-C using Xcode and platform SDKs

Skills

Application & Product Security: Threat modeling, security architecture/design review, secure code review, secure SDLC, vulnerability triage/remediation, web/API security, OWASP Top 10, Burp Suite, OAuth 2.0/OIDC, JWT, RBAC, privacy/data protection

Cloud, Container & DevSecOps: AWS/EC2, Terraform, Infrastructure as Code (IaC), Docker/container hardening, software supply-chain security, CI/CD, GitHub Actions, SAST, SCA, secrets/container/IaC scanning, dependency review, SBOM; CodeQL, Dependabot, Gitleaks, Trivy, CycloneDX

Engineering: Java/Spring Boot, Spring Security, Keycloak, C++, Python, TypeScript, Linux; AI-assisted development/review with OpenAI Codex, GitHub Copilot, CodeRabbit

Languages: English (fluent), Russian (fluent), Spanish (fluent), Portuguese (fluent), German (conversational)

Education

University of Illinois at Urbana-Champaign

Champaign, IL, USA

GPA: 4.0/4.0

Master of Science in Computer Science, Machine Learning Specialization

Jan 2017 - Dec 2017

Bachelor of Science in Computer Science, Bronze Tablet Honoree

Aug 2013 - Dec 2016